

Six Birds Foundations VI: A Catalog of Dynamical Structural Laws

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Abstract

A static structural law classifies fixed data. Many questions about dynamics concern runs instead: whether every orbit eventually descends, whether a run terminates while its presentation changes, whether a cover must move with its target. This paper states thirteen candidate laws of this kind and sorts them by what is proved. Seven are laws, each with a concrete system in which the difficult hypothesis is proved (for one of them only a small arithmetic example); two are general theorems with imported classical instances; four are templates, whose implication is proved while the difficult hypothesis remains an open obligation. Lean coverage is reported separately. The concrete results below are all checked in Lean. For the accelerated Collatz map every orbit satisfies an exact affine ledger, and at every depth m two starting values agree modulo 2^m but have different next division exponents, so no fixed residue class predicts the ledger. Under binary reverse-and-add, the orbit of 22 enters a four-phase family and never reaches a palindrome. The Ducci map on binary vectors of length 2^k is nilpotent. For move systems whose distinct legal moves commute, started where no infinite legal run exists, any two legal runs that end in stable states end in the same state with the same firing counts. Classical results are imported with attribution, nothing is labelled new, and the Collatz, Erdős–Straus, Lychrel and Recamán problems remain open.

Keywords. emergence calculus; dynamical structural law; liveness; template; Collatz affine ledger; split pair; reverse-and-add; Ducci map; abelian sandpile; odometer; Six Birds Theory.

1 Introduction

A structural law, in the sense this series gives the term, is a theorem about declared abstract data — a carrier, a quotient, a ledger, a family of patches — so that any system supplying the data inherits the conclusion. Foundations IV collected fifty-two such laws [51]. Every one of them is about fixed data: one map, one pair of quotients, one declared finite family of patches. None quantifies over the steps of a run. Yet many of the questions that make dynamics hard are questions about runs. Does every orbit of a map eventually descend? Does a run terminate when the way its states are written down changes at every step? Can a family of targets be covered when no fixed finite package of patches suffices? What happens when the obstruction to progress is produced by the run’s own history, or chosen by an adversary? When do route choices integrate into a conserved ledger, when do local defects turn into motion, what survives repeated loss, and how do local rules force global structure?

This paper states thirteen candidate laws, G1–G13, one for each of these questions, and sorts them by what has actually been proved. Each entry has a general statement whose *hard hypothesis*

— a coverage property, a closed family of states, a hierarchy of forced scales — carries the difficult content. Seven entries are *laws*: for each, the hard hypothesis is proved here for a concrete system, although for G13 that system is only a small arithmetic example. Two are *general theorems*, whose hypotheses are ordinary data and whose classical instances are imported. Four are *templates*: the implication is proved, but its hard hypothesis is not established in this paper for any system of interest and remains an open obligation. Table 1 records the sorting, and separately what is checked in Lean.

1.1 A worked example: residues do not predict the Collatz ledger

The accelerated Collatz map sends an odd number n to $T(n) = (3n + 1)/2^{a(n)}$, where $a(n)$ is the exponent of the largest power of 2 dividing $3n + 1$. Take $n = 5$ and $n = 9$. Both are congruent to 1 modulo 4, so they agree in their last two binary digits. But $3 \cdot 5 + 1 = 16$ gives $a(5) = 4$, while $3 \cdot 9 + 1 = 28$ gives $a(9) = 2$. Knowing n modulo 4 does not determine even the first division exponent.

The same happens at every depth: for each $m \geq 1$ there are two odd numbers that agree modulo 2^m whose first division exponents are m and at least $m + 1$ (Theorem 6.3). The pair above is the case $m = 2$. At the same time the bookkeeping of an orbit is exact. After k steps, with A_k the sum of the first k exponents,

$$2^{A_k} n_k = 3^k n + B_k,$$

for an explicit integer B_k determined by the exponents (Theorem 6.2). The orbit has dropped below its start at step k exactly when $2^{A_k} n > 3^k n + B_k$, an inequality between integers. Whether every orbit eventually satisfies it is the Collatz conjecture, which this paper does not settle. The two facts show where the difficulty lives: in an exact ledger whose next entry is not predicted by any fixed number of low-order digits.

1.2 Obstructions at every depth

Two notions recur throughout the paper. A *quotient* q_m records what an observer sees at depth m , for example the last m binary digits of a number, and a *readout* r records a future fact, for example the next division exponent. A *split pair* for q_m and r is a pair of states with the same quotient and different readouts. If a split pair exists at every depth, no observer who looks at a fixed number of digits can predict the readout.

The Collatz pairs above are such an obstruction for the last m binary digits and the first division exponent. The orbits of $2^L - 1$ give a second one for the time of first descent: they lie in the residue class of -1 modulo 2^m once $L \geq m$, and they stay above their start for at least $L - 1$ steps (Remark 6.5). So no bound on the time of first descent can depend only on the residue of n modulo a fixed power of 2.

Each entry of the catalog is stated in one form: the data of a run, a certificate in the run's own terms, and the named hypothesis that makes the certificate available. When that hypothesis is not yet proved for any system of interest, the entry is a *template*, and its value is that it names the obligation precisely and lets it be recognized in unrelated systems. The split-pair condition above, for instance, is exactly what must be avoided by any proof that a fixed-depth reading decides a run; the same condition reappears for binary reverse-and-add, where a different tool, a family of states closed under the map and free of palindromes, settles one orbit (Result B below).

1.3 Main results

The following four results are proved in the text and checked in Lean. Their relation to the catalog differs.

- Result A (Section 6.1) gives the certificate that G1 asks for, in exact integer form, and shows that no fixed number of binary digits predicts it. It does not establish the hypotheses of G1.
- Result B (Section 4.3) establishes the hard hypothesis of G5 for one orbit.
- Result C (Section 4.5) is a collapse example for G10. It shows that parity cannot be the protected invariant for vectors of length 2^k .
- Result D (Section 4.4) is the general statement of G8, with two examples that separate it from weaker properties.

Theorem (A: the Collatz ledger and its split pairs). *Every orbit of the accelerated Collatz map satisfies $2^{A_k}n_k = 3^k n_0 + B_k$ for every k . For every depth $m \geq 1$ there are odd positive $n_1 \equiv n_2 \pmod{2^m}$ whose first division exponents are at least $m + 1$ and exactly m . Hence no fixed residue class modulo 2^m determines the ledger (Theorems 6.2 and 6.3).*

Theorem (B: reverse-and-add in base 2). *Under binary reverse-and-add, the orbit of $22 = 10110_2$ reaches a four-word family after four steps, cycles through its four phases with a parameter that grows by one per cycle, and never reaches a palindrome (Theorem 4.6 and Corollary 4.7).*

Theorem (C: Ducci nilpotency). *On cyclic binary vectors of length 2^k , the map $x_i \mapsto x_i + x_{i+1}$ over $\mathbb{F}_2$ vanishes after 2^k steps (Theorem 4.18).*

Theorem (D: abelian move systems). *Consider a system of local moves in which any two distinct moves that are legal together remain legal after each other and commute. Start from a configuration that admits no infinite legal run. Then any two legal runs that end in stable configurations end in the same configuration, and they fire every site the same number of times (Theorem 4.11). Two small examples show that a unique final configuration does not give canonical firing counts, and that termination does not give a unique final configuration (Examples 4.13 and 4.14).*

1.4 What is imported, and what is not claimed

The deep theorems the catalog relies on are imported with attribution: Goodstein’s theorem and the Kirby–Paris independence result, Dhar’s abelian property and the least-action principle of Fey, Levine and Peres, the solution of the angel problem, the aperiodic tilings of Berger and Robinson, the de Bruijn–Erdős compactness theorem, and the density theorems of Bourgain and Kontorovich together with the reciprocity obstruction of Haag, Kertzer, Rickards and Stange. The Collatz conjecture, the Erdős–Straus conjecture, the base-10 Lychrel problem, Recamán’s coverage question, the Langton’s-ant highway conjecture, Gilbreath’s conjecture and the chromatic number of the plane all remain open.

Roadmap. Section 2 places the paper in the literature, and Section 3 fixes the vocabulary and gives the catalog. Section 4 treats the laws, Section 5 the general theorems and Section 6 the templates; the four headline results appear in their entries. Section 7 lists inferences the results do not support, and Section 8 records what is formalized. The supplement holds the source concordance, the full register of non-implications, the case analyses and instance lists, the finite computations, the reading of three laws as asymptotic statuses, and the replay commands.

2 Related work

The Six Birds series. Foundations IV is the static catalog this paper extends. It presents fifty-two laws in a six-part form — setting, statement, proof, case analysis, relation to the six structural roles, and limits — each about fixed data [51]. What separates each entry here from Foundations IV is a quantifier over an unbounded run, scale or orbit. Foundations V studies systems that carry out their own closure and repair [48]; the laws here are certified from outside, by an analyst auditing a run, and no law credits a system with certifying itself. The quotient vocabulary of Section 3, including split pairs and future-sufficient quotients, is taken from Foundations IV and from the paper on holonomy with memory [53, 52]. Currencies as constraints, ledgers, potentials and shadow prices come from Tsiokos [50], obstructions that survive every native probe from Tsiokos [54], and the asymptotic statuses used in Supplement Section S4 from the paper on audited operational realisability [49].

Classical sources. Each law with a classical core imports it. Termination by a decreasing assignment into a well-founded order is Floyd’s method [20]; Goodstein’s theorem and its unprovability in Peano arithmetic are due to Goodstein [23] and Kirby and Paris [27]. The potential method of amortized analysis is Tarjan’s [47, 12]. The greedy expansion into unit fractions goes back to Fibonacci and Sylvester [45]. The base-2 reverse-and-add result is recorded in the OEIS and on MathPages [37, 7]; we know of no refereed proof, and the one in Section 4.3 is self-contained. The angel problem was settled by Máthé [31], Kloster [28], Bowditch [5] and Gács [22]. Order independence for sandpiles is Dhar’s abelian property [16], and the least-action principle is due to Fey et al. [19]; abelian networks generalize both [26]. The traffic automaton rule 184 is analyzed by Fuké [21]. Ducci sequences are treated by Chamberland and Thomas [9] and Breuer et al. [6]. Aperiodicity by hierarchical forcing is due to Berger [1] and Robinson [41]. The de Bruijn–Erdős theorem is from de Bruijn and Erdős [13], and the lower bound 5 for the chromatic number of the plane from de Grey [14]. Thin orbits are treated by Bourgain and Fuchs [3] and Bourgain and Kontorovich [4], and the reciprocity obstruction by Haag et al. [24]. For the Collatz problem we rely on the surveys of Lagarias [29, 30] and on Tao [46].

Sources of the laws. Each law isolates the structure behind a problem with a small rule and a global obstruction: a tempting argument that relies on an auxiliary object, and the certificate in the original system that would make the argument valid. Supplement Section S8 works this out for Collatz (G1) and sandpiles (G8) and records two related patterns that are not laws of this paper.

3 Setting

Runs. A *run* on a carrier X is a sequence $x_0, x_1, x_2, \dots$ with $x_{k+1} = T(x_k)$ for an update $T : X \rightarrow X$, or $x_{k+1} = T_k(x_k)$ when the update may change with the step. A set $A \subseteq X$ of terminal states is declared when the question is whether a run stops. The laws of this paper quantify over whole runs: over every step, every legal order of moves, every scale, or every orbit. That unbounded quantifier is what separates each law from its nearest relative in Foundations IV, whose laws are about fixed data.

Quotients, descent and split pairs. A *quotient* is a map $q : X \rightarrow Q$ that records what an observer can see. A map F observed through a readout r *descends* through q when $r \circ F$ is a function of q , that is, when there is no *split pair*: no x, x' with $q(x) = q(x')$ and $r(F(x)) \neq r(F(x'))$.

[51]. When a whole family of future readouts $r_j \circ F^j$ is considered, q is *future-sufficient* if every one of them descends. Several laws are step-indexed versions of this test: a fixed quotient that fails to be future-sufficient is exhibited by a split pair at some future step.

Hard hypotheses. Each entry G1–G13 has a general statement. Its *hard hypothesis* is the hypothesis that carries the difficult content in applications — a coverage property, a closed family of states, a hierarchy of forced scales — as opposed to routine data such as a map or an order.

Laws, general theorems and templates. Each entry receives one of three statuses, decided by the mathematics.

- A *law* is a general statement together with a concrete system in which its hard hypothesis is proved in this paper, so that the conclusion is established for that system.
- A *general theorem* is a statement whose hypotheses are ordinary data that can be checked system by system; its classical instances rest on imported results.
- A *template* is an implication $H \Rightarrow C$ whose hard hypothesis H is not established for any system of interest. H is the *open obligation* of the template. A template says nothing about a particular system until H has been established for it.

A statement is a *reformulation* when it is an equivalence obtained by unfolding definitions. Which statements are also checked in Lean is a separate question, recorded in the last column of Table 1 and in Section 8.

Instances. An instance of an entry is a concrete system with its data. An instance is proved here, imported from a cited theorem, or open.

Certificates. Several entries concern a statement about an extended or auxiliary object — the 2-adic integers, an ordinal, a potential — that is used to settle a question about the original system. What may be concluded about the original system is a *certificate*: a checkable statement in its own terms, such as an inequality between integers or the fact that a run has stopped. The auxiliary object itself is never treated as a state of the original system.

The catalog at a glance. Table 1 lists the thirteen entries in the order in which they are treated. The first column names each entry with a one-line gloss and the section where it is treated. The second gives its status. The third names the instance proved in this paper for a law, and the open obligation of a template, with the principal instances otherwise. The fourth says what is checked in Lean. The full instance lists and case analyses are in Supplement Section S3.

Table 1. The catalog.

Entry	Status	Instance or open obligation	Checked in Lean
<i>Laws</i> (Section 4)			
G3 Amortized Potential Currency: a potential bounds total cost over every finite stretch (Section 4.1)	law	binary counter (Example 4.2); doubling array	telescoping identity and bound over integer costs; binary counter

Entry	Status	Instance or open obligation	Checked in Lean
G4 Moving-Cover Exhaustion: covers that change with the target close an infinite family (Section 4.2)	law	initial intervals (Example 4.4); greedy unit fractions; Erdős–Straus open	general statement; initial intervals
G5 Carry-Horizon Confinement: a closed family of states that avoids the target confines the run (Section 4.3)	law	orbit of 22 in base 2 (Corollary 4.7); 196 in base 10 open; no-go template	confinement; four-phase family; orbit of 22; no-go template
G8 Odometer Abelianization: commuting moves give an order-independent end state and firing counts (Section 4.4)	law	sandpiles (Example 4.12)	order independence; the two separating examples; a three-site sandpile; not general graphs
G10 Lossful Boundary Persistence: a relation survives every iterate of a lossy map if an invariant is regenerated (Section 4.5)	law	checksum cascade (Example 4.17); Ducci collapse; Gilbreath open	general statement; checksum cascade; Ducci over $\mathbb{F}_2$; integer Ducci for $n = 4$
G12 Finite Witness Radiation: a finite obstruction forbids a global colouring, and so do its images (Section 4.6)	law	Moser spindle, $\chi(\mathbb{R}^2) \geq 4$ (Example 4.21); $\chi(\mathbb{R}^2) \geq 5$ imported	general statement and the equivalence given compactness; the spindle graph has no 3-colouring; not the plane embedding
G13 Thin-Orbit Saturation: a sparse orbit misses only a density-zero part of a dense target (Section 4.7)	law, weak instance	powers of two, an arithmetic example (Example 4.24); thin-orbit saturation a template	density lemma; powers of two
<i>General theorems</i> (Section 5)			
G2 Transfinite Escrow: a value in a well-founded order that falls at every step forces termination (Section 5.1)	theorem	Goodstein sequences and Kirby–Paris (imported)	general statement, without extracting the terminal stage
G7 Adversarial Mobility Confinement: escape and confinement strategies against an adversary (Section 5.2)	theorem	the angel problem (imported)	general statement
<i>Templates</i> (Section 6)			
G1 Hidden Amortized Solvency: every nonterminal point eventually earns a certificate of descent (Section 6.1)	template	(H1) and (H2) of Template 6.7; for Collatz both open	reformulation; template; Collatz ledger and split pairs
G6 Endogenous Needle Generation: obstructions made by the run’s own history are covered or leave holes (Section 6.2)	template	bridge from rate conditions to the premises of Proposition 6.8; Recamán open	the two final steps

Entry	Status	Instance or open obligation	Checked in Lean
G9 Defect-Evacuation-to-Transport: local defects resolve or turn into certified motion (Section 6.3)	template	(H1) of Template 6.10; evacuation of every low-density rule-184 ring	census bound; packaging step; rule-184 partial results, car conservation, evacuation for $n \leq 8$
G11 Local-Rule Global-Anti-Symmetry: local rules with a forced hierarchy exclude every period (Section 6.4)	template	a hierarchy for a given tile set; Robinson and others (imported)	the implication

Map of results. Table 2 lists the numbered results. Each is proved where it is stated.

Table 2. Map of results.

Result	What it establishes
Theorem 4.1	Total cost over a finite stretch equals amortized cost plus the drop in potential, so amortized costs at most B give total cost at most $nB + \Phi(s_0)$.
Theorem 4.3	Patches that prove the target on their regions and cover every target prove it everywhere; under (H4) no finite list from the restricted class covers.
Theorem 4.5	A run that enters a closed pattern containing no target stays in it and never reaches a target.
Theorem 4.6	Binary reverse-and-add maps the four words $P_0(r), \dots, P_3(r)$ cyclically into the next family, and none is a palindrome.
Corollary 4.7	The orbit of 22 in base 2 enters the four-phase family and never reaches a palindrome.
Theorem 4.11	If legal moves commute and no infinite legal run exists, all stabilizing runs end in the same configuration with the same counter vector.
Theorem 4.16	A relation carried through one lossy step by a regenerated invariant persists through every iterate.
Theorem 4.18	Over $\mathbb{F}_2$ the Ducci map is nilpotent on words of length 2^k .
Theorem 4.20	A finite witness rules out a k -colouring, and its image under any map preserving E is again a witness.
Proposition 4.22	Under compactness, no k -colouring exists exactly when a finite witness does.
Theorem 4.23	A density-zero set is density zero relative to any target of positive lower density.
Theorem 5.1	A value in a well-founded order that falls at every step outside the terminal set forces termination.
Theorem 5.2	An invariant set closed under the angel's strategy and every legal devil reply certifies escape; the dual certifies confinement.
Theorem 6.2	After k odd steps of the accelerated Collatz map, $2^{A_k} n_k = 3^k n + B_k$.
Theorem 6.3	For each m , two odd numbers congruent modulo 2^m have different division exponents.
Corollary 6.4	The division exponent is not a function of n modulo 2^m for any m .
Proposition 6.6	The system is live exactly when no point lies in every bad tail N_k .
Proposition 6.8	Holes that are eventually reduced are eventually removed, and an unvisited point that stays unvisited is never visited.
Theorem 6.9	A nonincreasing count of unresolved defects is bounded.
Proposition 6.11	Rule 184 keeps jam-free rings jam-free and conserves cars; on rings of 2 to 8 cells, every configuration with at most half the cells occupied is jam-free after n steps.

4 Laws with instances

Each of the seven entries in this section has a general statement and a concrete system in which its hard hypothesis is proved, in the text below. Hypotheses are labelled (H1), (H2), ... within each entry.

4.1 G3: Amortized Potential Currency

Name. A potential acts as a *currency*: credit stored in the state pays for expensive steps later.

Let $s_0 \rightarrow s_1 \rightarrow \dots \rightarrow s_n$ be a finite stretch of a run in which step i has cost a_i , and let $\Phi \geq 0$ be a potential on states. The *amortized cost* of step i is $\hat{a}_i = a_i + \Phi(s_i) - \Phi(s_{i-1})$.

Theorem 4.1 (Amortized potential). *For every finite stretch,*

$$\sum_{i=1}^n a_i = \sum_{i=1}^n \hat{a}_i + \Phi(s_0) - \Phi(s_n) \leq \sum_{i=1}^n \hat{a}_i + \Phi(s_0).$$

If (H1) $\hat{a}_i \leq B$ at every step, then $\sum_{i=1}^n a_i \leq nB + \Phi(s_0)$.

Proof. The potential terms in $\sum \hat{a}_i$ telescope, and $\Phi(s_n) \geq 0$. □

This is the potential method of amortized analysis [47, 12]. G3 is a safety statement about every finite stretch and says nothing about eventual success; it assumes the potential rather than constructing it. A potential counts as a certificate only when it is nonnegative, enters through the exact telescoping formula, and bounds the cost claim actually made. A statistic that merely correlates with cost does not.

Example 4.2 (Binary counter). Let the state be a binary counter, let an increment cost the number of bits it changes, and let Φ be the number of 1 bits. An increment that turns r trailing ones into zeros and one zero into a one costs $r + 1$ and changes Φ by $1 - r$, so its amortized cost is exactly 2 and (H1) holds with $B = 2$. Hence n increments from zero change at most $2n$ bits, counting a newly created leading bit as a changed bit.

Figure 1 shows the first eight increments.

n	counter $n - 1 \rightarrow n$	bit positions flipped	cost a_n	Φ before $\rightarrow$ after	$\Delta\Phi$	$a_n + \Delta\Phi$
1	0000 $\rightarrow$ 0001	{0}	1	0 $\rightarrow$ 1	+1	2
2	0001 $\rightarrow$ 0010	{0, 1}	2	1 $\rightarrow$ 1	0	2
3	0010 $\rightarrow$ 0011	{0}	1	1 $\rightarrow$ 2	+1	2
4	0011 $\rightarrow$ 0100	{0, 1, 2}	3	2 $\rightarrow$ 1	-1	2
5	0100 $\rightarrow$ 0101	{0}	1	1 $\rightarrow$ 2	+1	2
6	0101 $\rightarrow$ 0110	{0, 1}	2	2 $\rightarrow$ 2	0	2
7	0110 $\rightarrow$ 0111	{0}	1	2 $\rightarrow$ 3	+1	2
8	0111 $\rightarrow$ 1000	{0, 1, 2, 3}	4	3 $\rightarrow$ 1	-2	2

Figure 1. Eight increments from zero, with bit positions numbered from the least significant bit. The displayed potential is the number of 1 bits.

For a doubling array with $\Phi = \max(0, 2 \text{ size} - \text{capacity})$, every insertion has amortized cost at most 3 by the same kind of computation [12].

4.2 G4: Moving-Cover Exhaustion

Name. The cover of the target family must *move*: which patch covers a target depends on the target.

Let $\{P(t) : t \in \mathcal{T}\}$ be an infinite family of targets. A *patch* α comes with a region $R_\alpha \subseteq \mathcal{T}$ on which it claims to prove P .

Theorem 4.3 (Moving-cover exhaustion). *Consider the hypotheses*

- (H1) *every patch proves $P(t)$ for every t in its region;*
- (H2) *every target lies in the region of some patch;*
- (H3) *if P holds everywhere, the declared patches cover every target;*
- (H4) *for a declared restricted class of patches, every finite list from the class misses some target.*

Then: (H1) and (H2) give $P(t)$ for every t ; under (H1) and (H3), P holds everywhere exactly when the patches cover every target; and under (H4) no finite list from the restricted class covers every target.

Proof. The first claim applies (H1) to the covering patch. The forward direction of the second is (H3), and its converse is the first claim. The third is (H4) applied to the given list. $\square$

Each conclusion is close to its hypothesis: (H2) is total coverage, and (H4) already says that every finite list misses a target. The substance lies in establishing (H2) and (H4) for a system, usually by a residual for each target that shrinks as the cover grows.

Example 4.4 (Initial intervals). Let $\mathcal{T} = \mathbb{N}$, and let the patch with index a cover the interval $[0, a)$. For a target t the residual $\max(0, t + 1 - a)$ decreases strictly as a grows while it is positive, and vanishes at $a = t + 1$, so (H2) holds. A finite list of patches misses the largest of its endpoints, so (H4) holds for the class of all patches. Initial intervals therefore exhaust the natural numbers as their endpoints grow, although no finite collection of them covers every natural number.

The greedy expansion of a fraction $p/q \in (0, 1)$ into distinct unit fractions is the classical instance. With $m = \lceil q/p \rceil$ the remainder after subtracting $1/m$ is $(pm - q)/(qm)$, and $0 \leq pm - q < p$, so the numerator strictly decreases and the expansion ends [45]. A fixed finite set of denominators has finitely many subset sums, so no fixed package covers all targets; the greedy cover succeeds because its denominators depend on the target. For the Erdős–Straus conjecture ($4/n = 1/x + 1/y + 1/z$ for all $n \geq 2$), a solution for a prime yields one for its multiples, and no finite family of polynomial identities can cover the class $n \equiv 1$ modulo every prime because 1 is a quadratic residue [32]. The conjecture is open, and density results [55, 17] do not give it.

4.3 G5: Carry-Horizon Confinement

Name. Predicting digit dynamics can need digits from ever deeper positions, as a carry travels; the *horizon* of useful digits recedes. A closed family of states *confines* the run when prediction fails.

Let $R : X \rightarrow X$ be an update and let a set of *targets* be declared; for reverse-and-add the targets are the palindromes.

Theorem 4.5 (Confinement by a closed pattern). *Let $\mathcal{P} \subseteq X$ satisfy (H1) $R(\mathcal{P}) \subseteq \mathcal{P}$ and (H2) no element of $\mathcal{P}$ is a target. If $R^N(x) \in \mathcal{P}$, then $R^{N+t}(x) \in \mathcal{P}$ and $R^{N+t}(x)$ is not a target for every $t \geq 0$.*

Proof. Induction on t . $\square$

The instance: reverse-and-add in base 2

A *binary word* here is a word over $\{0, 1\}$ that begins with 1; every positive integer has exactly one. For a word w let $\bar{w}$ be its reversal, which may begin with zeros. Binary reverse-and-add sends w to the binary word of $w + \bar{w}$, reading both as numbers; we write R for this map on words and on the positive integers they represent. A word is a *palindrome* when it equals its reversal. For $r \geq 2$ define the four words

$$\begin{aligned} P_0(r) &= 10 1^r 01 0^r, & P_1(r) &= 11 0^{r-2} 1 000 1^{r-2} 01, \\ P_2(r) &= 10 1^r 01 0^{r+1}, & P_3(r) &= 11 0^r 10 1^{r-1} 01, \end{aligned}$$

where b^j denotes j copies of the digit b .

Theorem 4.6 (Four-phase family). *For every $r \geq 2$, as binary words,*

$$R(P_0(r)) = P_1(r), \quad R(P_1(r)) = P_2(r), \quad R(P_2(r)) = P_3(r), \quad R(P_3(r)) = P_0(r+1),$$

and none of the four words is a palindrome.

Proof. Put $x = 2^r$. The value of each word and of its reversal is a polynomial in x with integer coefficients, given in Table 3. In each row the value and the reversal add up to the value of the next word; for the last row, $P_0(r+1)$ has value $12(2x)^2 - 3(2x) = 48x^2 - 6x$. Since a binary word is determined by its value, each identity of values is an identity of words. Finally, P_0 and P_2 begin with 1 and end with 0, and P_1 and P_3 begin with 11 and end with 01, so none equals its reversal. $\square$

Table 3. Values of the four phases and their reversals, with $x = 2^r$.

Word	Value	Reversal	Sum	Next word
$P_0(r)$	$12x^2 - 3x$	$12x - 3$	$12x^2 + 9x - 3$	$P_1(r)$
$P_1(r)$	$12x^2 + 9x - 3$	$12x^2 - 15x + 3$	$24x^2 - 6x$	$P_2(r)$
$P_2(r)$	$24x^2 - 6x$	$12x - 3$	$24x^2 + 6x - 3$	$P_3(r)$
$P_3(r)$	$24x^2 + 6x - 3$	$24x^2 - 12x + 3$	$48x^2 - 6x$	$P_0(r+1)$

The family is recorded in the OEIS and on MathPages [37, 7]; the proof here is self-contained.

Corollary 4.7 (The orbit of 22). *The orbit of 22 under binary reverse-and-add begins*

$$10110 \rightarrow 100011 \rightarrow 1010100 \rightarrow 1101001 \rightarrow 10110100 = P_0(2),$$

thereafter runs through $P_1(2), P_2(2), P_3(2), P_0(3), \dots$, and never reaches a palindrome.

Proof. The first four steps are direct computations ($22 + 13 = 35$, $35 + 49 = 84$, $84 + 21 = 105$, $105 + 75 = 180$), and none of the four initial words is a palindrome. From $P_0(2)$ on, the family of Theorem 4.6 satisfies (H1) and (H2) of Theorem 4.5, so no later word is a palindrome. $\square$

In base 10 no such certificate is known: whether the orbit of 196 reaches a palindrome is open, and finite searches are evidence only [56].

Why a certificate of this kind is needed

Let q_m be a family of quotients — for reverse-and-add in base b , $q_m(n) = n \bmod b^m$ records the last m digits — and let r_j be a family of future readouts, such as the last digits after j steps.

Template 4.8 (No fixed prediction depth). *Assume*

- (H3) *for every depth m there are a step j and states y, z with $q_m(y) = q_m(z)$ and $r_j(y) \neq r_j(z)$.*

Then no depth m is future-sufficient for the declared readouts.

Proof. The pair y, z at depth m is a split pair for q_m . □

The template holds for any quotients and readouts. For the Collatz exponent, with $q_m(n) = n \bmod 2^m$ and the readout $a(n)$, (H3) is Theorem 6.3. Whether (H3) holds for base-2 reverse-and-add with a stated family of readouts is open; the confinement of 22 does not need it.

4.4 G8: Odometer Abelianization

Name. The *odometer* is the vector of firing counts; *abelianization* means that the order of commuting moves stops mattering.

Definition 4.9 (Move system). A *move system* consists of a set C of configurations, a set V of sites, and for each $v \in V$ a move $m_v : C \rightarrow C$ with a legality predicate L_v on C . A *legal run* from x is a finite sequence of sites $v_0, \dots, v_{\ell-1}$ with L_{v_i} holding at the configuration reached before step i . Its *counter vector* records how often each site occurs. A configuration is *stable* when no move is legal there.

Definition 4.10 (Abelian compatibility). The move system satisfies (H1) when for every configuration c and all *distinct* sites $v \neq w$ that are both legal at c : w is legal at $m_v(c)$, v is legal at $m_w(c)$, and $m_w(m_v(c)) = m_v(m_w(c))$.

Theorem 4.11 (Order independence). *Assume (H1) and (H2): no infinite sequence of legal moves starts at x . If two legal runs from x end in stable configurations, then they end in the same configuration and have the same counter vector.*

Proof. Induction on the length of the first run. If it is empty, x is stable and the second run is empty as well. Otherwise let v be its first site. The second run ends in a stable configuration, so v is not legal at its end. By (H1), firing any site other than v keeps v legal, so v occurs in the second run. Moving that occurrence step by step to the front uses (H1) at each swap, keeps the run legal, and changes neither its final configuration nor its counter vector. Both runs now begin with v , and the induction hypothesis applies to their remainders from $m_v(x)$. □

This is the abstract form of Dhar’s abelian property for sandpiles [16]. Under (H2) every maximal legal run from x is finite and ends in a stable configuration, so all of them end in one configuration x° with one counter vector u_x , the *odometer*.

Example 4.12 (Sandpiles). Let G be a finite connected graph with a distinguished sink s . A configuration assigns a number of chips $\eta(v) \geq 0$ to each $v \neq s$. A site v is legal when $\eta(v) \geq \deg(v)$, and toppling it sends one chip along each edge at v ; chips sent to s are lost. For distinct legal v, w , toppling v does not decrease $\eta(w)$, so w stays legal, and both orders subtract the same vector of chip changes; this is (H1). For (H2), suppose an infinite legal run exists. The number of chips never increases, so every $\eta(v)$ stays bounded. A site that topples infinitely often sends infinitely many chips

to each neighbour, which must then topple infinitely often too. By connectedness some neighbour of s topples infinitely often, and each such toppling loses a chip to s ; this contradicts the finiteness of the chip count. Theorem 4.11 therefore gives every sandpile a unique stable configuration and odometer.

One small case in which sites exchange chips has three mutually adjacent sites, each also joined to the sink, so that every degree is 3. Toppling a site removes three of its chips and adds one to each other site, and one chip is lost to the sink; so every toppling lowers the total by exactly one, which gives (H2) at once.

Figure 2 shows one toppling in this case.

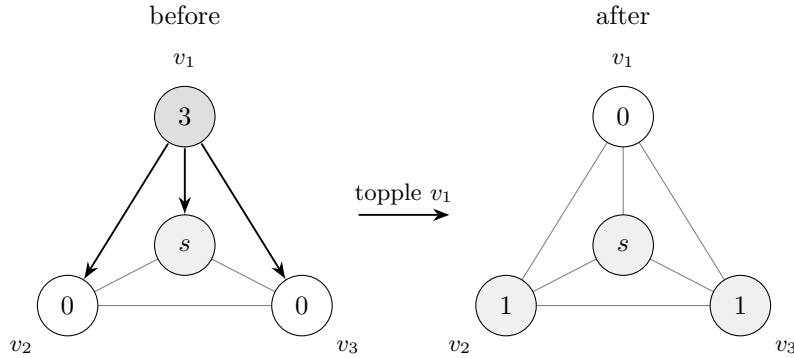


Figure 2. One toppling on the three-site graph: $(3, 0, 0) \rightarrow (0, 1, 1)$. The arrows show one chip sent along each edge from v_1 , including the edge to the sink s .

Rotor-routing belongs to the same family of abelian networks [26]. The next two examples separate the theorem from weaker properties of rewriting systems.

Example 4.13 (A unique final configuration without canonical counts). Take configurations S, A, B, N and legal moves $r : S \rightarrow A$, $s : S \rightarrow B$, $p : A \rightarrow N$, $t : A \rightarrow N$, $q : B \rightarrow N$, and no others. Every run terminates, and every maximal run ends at N . The three maximal runs rp , rt and sq have three different counter vectors. (H1) fails at S : after r the move s is no longer legal.

Example 4.14 (Termination without a unique final configuration). Take configurations S, A, B and legal moves $r : S \rightarrow A$ and $s : S \rightarrow B$ only. Both runs terminate, at the distinct stable configurations A and B .

Remark 4.15 (Least action). The least-action principle of Fey et al. [19] compares the odometer with every nonnegative script of firings that stabilizes the configuration, including scripts that pass through illegal intermediate configurations. It is imported. The weaker comparison with scripts that are themselves realized by legal stabilizing runs adds nothing under the hypotheses of Theorem 4.11, which make all such scripts equal.

4.5 G10: Lossful Boundary Persistence

Name. A boundary relation *persists* through a map that is *lossful*, that is, not injective.

Let $D : X \rightarrow X$ be a map, $b : X \rightarrow B$ a boundary readout, $\rho : B \rightarrow B$ a boundary update, and I a predicate on X , the interior invariant. The run is $x_{n+1} = D(x_n)$.

Theorem 4.16 (Persistence through iterated loss). *Assume*

- (H1) $I(x)$ implies $b(D(x)) = \rho(b(x))$, and
- (H2) $I(x)$ implies $I(D(x))$,

and $I(x_0)$. Then $I(x_n)$ and $b(x_n) = \rho^n(b(x_0))$ for every $n \geq 0$.

Proof. Induction on n . □

The theorem holds for every map D . It is of interest when D is not injective, so that interior information is lost while the boundary relationship survives because the invariant is regenerated at every step.

Example 4.17 (Checksum cascade). On $X = \{0, 1\}^2 \times \{0, 1\}$ let $D((a, b), c) = ((c, 0), c)$, with boundary $b((a, b), c) = c$, ρ the identity and invariant $c = a \oplus b$. The map is not injective, and (H1) and (H2) hold: if $c = a \oplus b$ then the new state $((c, 0), c)$ satisfies $c = c \oplus 0$. The checksum therefore persists forever while the two data bits are lost.

A collapse example: Ducci maps

For $n \geq 1$ let D act on cyclic binary vectors $x = (x_0, \dots, x_{n-1}) \in \mathbb{F}_2^n$ by $D(x)_i = x_i + x_{i+1}$, indices modulo n .

Theorem 4.18 (Ducci nilpotency over $\mathbb{F}_2$). *If $n = 2^k$, then $D^n(x) = 0$ for every $x \in \mathbb{F}_2^n$.*

Proof. Write $D = I + S$ with S the cyclic shift. Over $\mathbb{F}_2$, induction on j gives $D^{2^j}(x)_i = x_i + x_{i+2^j}$, since squaring $I + S^{2^j}$ gives $I + S^{2^{j+1}}$. For $j = k$ the shift $S^{2^k} = S^n$ is the identity, so $D^n(x)_i = 2x_i = 0$. □

The result is classical [9, 6]. For vectors of length 2^k the parity structure is destroyed completely, so no nonzero persistence claim for all such vectors can rest on parity.

Remark 4.19 (Integer Ducci tuples). The integer Ducci map sends $(x_0, \dots, x_{n-1})$ to $(|x_0 - x_1|, \dots, |x_{n-1} - x_0|)$. Modulo 2 it is D , and it commutes with multiplication by 2. For $n = 2^k$, Theorem 4.18 makes every entry even after n steps, after which the run is twice another Ducci run. Since the entries never exceed the largest entry of the first iterate, they become divisible by arbitrarily large powers of 2 and hence vanish. For $n = 4$ the argument gives an explicit bound: every $x \in \mathbb{N}^4$ reaches the zero tuple after $4m$ steps of the integer map, where m is the largest entry of x , because after $4k$ steps every entry is a multiple of 2^k and at most m , and $2^m > m$. For other lengths integer tuples can cycle without reaching zero, for example $(0, 1, 1) \rightarrow (1, 0, 1) \rightarrow (1, 1, 0) \rightarrow (0, 1, 1)$.

Gilbreath's conjecture, read as a persistence claim for the leading entry of iterated absolute differences of the primes, is open [34].

4.6 G12: Finite Witness Radiation

Name. A finite witness *radiates*: its images under the symmetries of the graph are witnesses too.

Let E be a graph relation on a set X . A *proper k -colouring* is a map $c : X \rightarrow \{1, \dots, k\}$ with $c(x) \neq c(y)$ whenever $E(x, y)$. A *finite witness* for k is a finite set $W \subseteq X$ whose induced graph has no proper k -colouring. A map $\gamma : X \rightarrow X$ *preserves E* when $E(x, y) \Leftrightarrow E(\gamma x, \gamma y)$.

Theorem 4.20 (Finite witnesses). *Let W be a finite witness for k .*

1. X has no proper k -colouring.

2. If γ preserves E , then $\gamma(W)$ is a finite witness for k .

Proof. A proper colouring of X restricts to one of W . A proper colouring c of $\gamma(W)$ pulls back to the proper colouring $c \circ \gamma$ of W , because γ sends edges to edges; γ need not be injective. $\square$

Example 4.21 (The Moser spindle). In the plane with $E(x, y)$ meaning $\|x - y\| = 1$, take two rhombi with vertices A, B, D, C and A, B', D', C' in cyclic order and unit sides, each made of two equilateral unit triangles (ABC , BCD and $AB'C'$, $B'C'D'$), so that A and D are opposite vertices with $\|A - D\| = \sqrt{3}$, and likewise $\|A - D'\| = \sqrt{3}$. Rotate the second rhombus about A by the angle θ with $2\sqrt{3}\sin(\theta/2) = 1$, which exists because $1 < 2\sqrt{3}$; then $\|D - D'\| = 1$. The seven points A, B, C, D, B', C', D' are distinct, and the eleven pairs $AB, AC, BC, BD, CD, AB', AC', B'C', B'D', C'D'$ and DD' are at unit distance. In a proper 3-colouring, B and C take the two colours other than that of A , so D takes the colour of A ; likewise D' does, and D, D' are adjacent. So the seven points form a finite witness for $k = 3$, and $\chi(\mathbb{R}^2) \geq 4$ by Theorem 4.20; by part 2, so does every rotated and translated copy.

Figure 3 shows the seven points and the eleven unit distances.

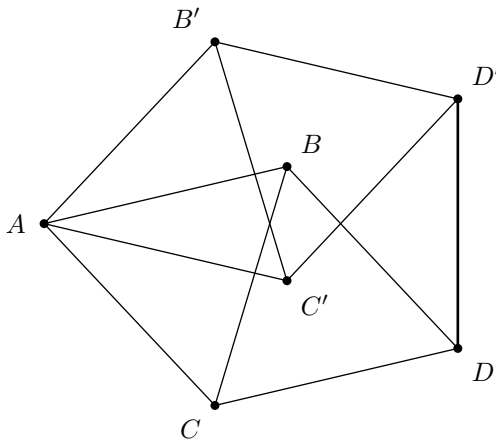


Figure 3. Two unit rhombi sharing A , with the second rotated by θ where $2\sqrt{3}\sin(\theta/2) = 1$. The heavier edge joins D and D' ; all eleven drawn edges have unit length.

Proposition 4.22 (Completeness of finite witnesses under compactness). *Assume*

- (H1) *if X has no proper k -colouring, then some finite witness for k exists.*

Then X has no proper k -colouring exactly when a finite witness for k exists.

Proof. The forward direction is the hypothesis and the backward direction is Theorem 4.20. $\square$

The hypothesis is the de Bruijn–Erdős theorem [13], which uses the axiom of choice. For the unit-distance graph of the plane the current bounds are $5 \leq \chi(\mathbb{R}^2) \leq 7$. The lower bound is a finite witness in exactly the sense of Theorem 4.20: de Grey constructed a 1581-vertex unit-distance graph that is not 4-colourable [14], and later work reduced this to 553 vertices [25] and to 509 [38]. The upper bound is the classical hexagonal colouring. The exact value is open, and questions of this kind can depend on the set-theoretic background [42, 18]; the finite lower bound does not, once the graph and its non-colourability are accepted.

4.7 G13: Thin-Orbit Saturation

Name. An orbit that is *thin* (sparse in a precise sense) can still *saturate* a target, missing only a vanishing proportion of it.

Let $m, t : \mathbb{N} \rightarrow \mathbb{N}$ count, up to N , the missed members and all members of a target set.

Theorem 4.23 (Relative density). *If $m(N) = o(N)$ and $t(N) \geq cN$ for all N with a fixed $c > 0$, then $m(N) = o(t(N))$.*

Proof. Given $\varepsilon > 0$, eventually $m(N) \leq \varepsilon cN \leq \varepsilon t(N)$. □

Example 4.24 (Powers of two). Let the target be the positive integers and let the represented set be the integers that are not powers of two. The missed family, the powers of two, is infinite, and its count up to N is at most $\lfloor \log_2 N \rfloor + 1$, so its relative density is zero by Theorem 4.23 with $t(N) = N$. The represented set has density one and still misses infinitely many targets.

This is a small arithmetic example of the counting phenomenon, not a thin-orbit result. The setting that motivates G13 is a group Λ acting on a discrete set, with an orbit projected to integers and observed against a target A . There Λ is *thin* (Zariski dense of infinite index in an arithmetic group) and A is *thick* (a finite union of residue classes, so of positive density).

Template 4.25 (Thin orbits saturate thick targets). *Assume that Λ is thin, that its congruence quotients have a uniform spectral gap, that A passes the local congruence conditions, and*

- (H1) *an imported theorem gives $m(N) = O(N^{1-\eta})$ for some $\eta > 0$.*

Then the proportion of missed admissible targets tends to zero.

Proof. (H1) gives $m(N) = o(N)$, and a thick target gives $t(N) \geq cN$ for all large N , which suffices for the argument of Theorem 4.23. The first three hypotheses are the setting in which such bounds are proved and are not used. □

Apollonian circle packings are the motivating instance. Curvatures of four mutually tangent circles satisfy Descartes' relation $k_1^2 + k_2^2 + k_3^2 + k_4^2 = \frac{1}{2}(k_1 + k_2 + k_3 + k_4)^2$, and the integral packing generated from $(-1, 2, 2, 3)$ has curvatures only in the residue classes $\{2, 3, 6, 11, 14, 15, 18, 23\}$ modulo 24. Bourgain and Fuchs proved that the curvatures have positive density [3], and Bourgain and Kontorovich that almost every admissible integer occurs, with $O(N^{1-\eta})$ exceptions [4]. Haag, Kertzer, Rickards and Stange showed that the full local-to-global conjecture is nevertheless false for many packings: certain quadratic and quartic families pass the congruence conditions and are still missed, for reasons of quadratic reciprocity [24]. Those families have density zero, so they do not contradict the template; they are a second kind of exception, distinct from failing a congruence condition, and Example 4.24 is the same pattern in miniature.

5 General theorems

The two entries in this section are theorems whose hypotheses are ordinary data, checked system by system. Their classical instances rest on imported results.

5.1 G2: Transfinite Escrow

Name. The *escrow* is a value, held in a well-founded order, that must fall at every step; its fall pays for termination. The presentation may change at every step, and the escrow is compared across the change.

Let $x_0, x_1, \dots$ be a run with $x_{k+1} = T_k(x_k)$ and terminal set A . At each step a *presentation* $\pi_k : X \rightarrow P_k$ records how the state is written, and an *escrow* $e_k : P_k \rightarrow W$ assigns it a value in a well-founded order $(W, <)$.

Theorem 5.1 (Transfinite escrow). *Assume $(W, <)$ is well-founded and*

- (H1) *whenever $x_k \notin A$, $e_{k+1}(\pi_{k+1}(T_k(x_k))) < e_k(\pi_k(x_k))$.*

Then some x_k lies in A .

Proof. Otherwise the escrow values $e_k(\pi_k(x_k))$ would form an infinite strictly decreasing sequence in W . $\square$

This is Floyd's ranking principle [20]. The condition compares the escrow before a step with the escrow after the step *and* the change of presentation, checked as one transition. A decrease that holds only before or only after an unrecorded change of presentation does not discharge it.

Goodstein sequences are the classical instance. In base k the state n is written in hereditary base k ; one step rewrites it in base $k+1$ and subtracts 1. Replacing every base symbol by ω gives an ordinal escrow below ε_0 that is unchanged by the rewriting and strictly decreased by the subtraction. From $4 = 2^2$, rewriting gives $3^3 = 27$ and subtracting gives $26 = 2 \cdot 3^2 + 2 \cdot 3 + 2$: the number grows while the escrow drops from ω^ω to $\omega^2 \cdot 2 + \omega \cdot 2 + 2$. Every Goodstein sequence therefore terminates [23], and by the theorem of Kirby and Paris [27] this cannot be proved in Peano arithmetic.

5.2 G7: Adversarial Mobility Confinement

Name. The angel's *mobility* is set against the devil's power to *confine* it.

In the angel game on $\mathbb{Z}^2$ with the Chebyshev distance, an angel of power p moves each turn to an undeleted square within distance p , and the devil then deletes at most b squares. The angel wins if it can move forever. The theorem below concerns any game of this shape. A state consists of the angel's position and a visible obstruction record. At each turn the angel moves to a position that a legality predicate allows, and the devil then replaces the record by one that a second legality predicate allows. A strategy is a function of the current state. The angel is *stuck* at a state with no legal move.

Theorem 5.2 (Escape and confinement certificates). *Fix a game of this shape.*

1. *Let σ_A be an angel strategy and I a set of states such that at every state in I the move of σ_A is legal and every legal devil reply to it leads to a state in I . Let σ_D be a devil strategy whose reply to every legal move is legal. Then the play of σ_A against σ_D from any state in I stays in I , and the angel is never stuck.*
2. *Let σ_D be a devil strategy and r a natural-number rank on states such that the reply of σ_D to every legal angel move is legal and strictly lowers r . Then against every angel strategy that makes a legal move whenever one exists, the angel is stuck at some turn $n \leq r(s_0)$, where s_0 is the initial state.*
3. *No game has an angel strategy σ_A with a set I as in part 1, a devil strategy σ_D with a rank r as in part 2, and a state in I .*

Proof. Parts 1 and 2 are inductions on turns, the second bounded by the rank. For part 3, play σ_A against σ_D from a state in I : by part 1 the angel moves legally at every turn, and each reply lowers the rank, which cannot happen forever. $\square$

The theorem checks certificates; it does not produce strategies. Without a state in I part 3 fails, since an empty set satisfies the hypothesis of part 1 vacuously. For the standard game with $b = 1$ the answer is known: the angel of power 1 loses [2, 11], and an angel of power 2 wins [31, 28], with further proofs for power 4 and high power [5, 22].

6 Templates

Each of the four entries in this section is a template: an implication whose hard hypothesis is not established, in this paper, for any system of interest. The open obligation is named in each case, and each entry is preceded by what is proved about its principal system. Hypotheses are labelled (H1), (H2), ... within each entry.

6.1 G1: Hidden Amortized Solvency

Name. A point is *solvent* once it has earned a certificate of descent; the debt it carries before that is *hidden* in the gap between the orbit and its start, and paid off *amortized* over many steps.

The concrete system: the Collatz ledger

Definition 6.1 (Accelerated Collatz map). For a natural number n let $a(n)$ be the exponent of the largest power of 2 dividing $3n + 1$, and let $T(n) = (3n + 1)/2^{a(n)}$. For a starting value n write $n_j = T^j(n)$ and $a_j = a(n_j)$, and define $A_k = \sum_{j < k} a_j$ together with $B_0 = 0$ and $B_{j+1} = 3B_j + 2^{A_j}$.

On odd positive integers T is the usual accelerated map: $T(n)$ is again odd and positive, and $T(1) = 1$. The definition makes sense for every natural number, and the statements below hold there; the odd positive integers are the case of interest.

Theorem 6.2 (Affine ledger). *For every natural number n and every $k \geq 0$,*

$$2^{A_k} n_k = 3^k n + B_k.$$

Proof. For $k = 0$ both sides equal n . By definition $2^{a_k} n_{k+1} = 3n_k + 1$, so

$$2^{A_{k+1}} n_{k+1} = 2^{A_k} (3n_k + 1) = 3(3^k n + B_k) + 2^{A_k} = 3^{k+1} n + B_{k+1}. \quad \square$$

The ledger is standard in the Collatz literature [29].

For $n > 0$ the ledger turns descent into an inequality between integers: $n_k < n$ exactly when $2^{A_k} n > 3^k n + B_k$. This is the certificate that G1 asks for.

Theorem 6.3 (Split pairs). *Let $m \geq 1$, let $c \in \{1, 2\}$ satisfy $2^{m+1}c \equiv 1 \pmod{3}$, and put*

$$n_1 = \frac{2^{m+1}c - 1}{3}, \quad n_2 = n_1 + 2^m.$$

Then $n_1 \equiv n_2 \pmod{2^m}$, while $a(n_1) \geq m + 1$ and $a(n_2) = m$.

Proof. The congruence is immediate. By construction $3n_1 + 1 = 2^{m+1}c$, so 2^{m+1} divides $3n_1 + 1$. Further $3n_2 + 1 = 2^{m+1}c + 3 \cdot 2^m = 2^m(2c + 3)$, and $2c + 3$ is odd. $\square$

Both numbers are odd and positive, since $3n_1 + 1$ is even and $n_1 \geq 1$. For $m = 1$ the pair is $(1, 3)$; for $m = 2$ it is $(5, 9)$, the pair of Section 1.1.

Corollary 6.4. *For no $m \geq 1$ is the division exponent $a(n)$ of an odd positive n a function of n modulo 2^m . In the language of Section 3, no quotient $n \mapsto n \bmod 2^m$ is future-sufficient for the ledger.*

Proof. The pair of Theorem 6.3 consists of odd positive numbers, as noted after it; they agree modulo 2^m and have different division exponents. $\square$

Remark 6.5 (Three further facts). 1. In the 2-adic integers, -1 is a fixed point: $3(-1) + 1 = -2$ has exponent 1. It is not an integer orbit, but integer orbits can imitate it for a long time.

2. For $n = 2^L - 1$ with $L \geq 2$, induction gives $n_i = 3^i 2^{L-i} - 1$ and $a_i = 1$ for $0 \leq i \leq L - 2$. The orbit therefore stays above its start for at least $L - 1$ steps, so no bound on the first descent time that is uniform in n exists.

3. If an orbit returns to its start after k steps, then $(2^{A_k} - 3^k)n = B_k$. A cycle of positive integers can therefore occur only for exponent sequences that make $B_k/(2^{A_k} - 3^k)$ a positive integer.

The template

Let $T : X \rightarrow X$ be an update with a set A of terminal states, and let $<$ be a declared relation on X , usually an order. A point x *descends at step k* when $T^k(x) < x$. For $k \geq 1$ the *bad-tail set* N_k consists of the points $x \notin A$ that descend at no step j with $1 \leq j \leq k$. The system is *live* when every $x \notin A$ descends at some step.

Proposition 6.6 (Bad-tail reformulation). *The system is live exactly when no point lies in every N_k :*

$$(\forall x \notin A \exists k \geq 1 : T^k(x) < x) \iff \bigcap_{k \geq 1} N_k = \emptyset.$$

Proof. A point in every N_k never descends, and a point that never descends lies in every N_k . $\square$

This only restates liveness. The content of G1 is in what could empty the intersection. A point in every N_k has an infinite orbit that never descends. Suppose the system maps into a larger space $\hat{X}$ — for Collatz, the 2-adic integers — whose non-descending fixed or periodic points outside X form a *ghost set* Γ , and let $U_1, U_2, \dots$ be subsets of $\hat{X}$, typically shrinking neighbourhoods of Γ . The template below uses only the sets U_h , not Γ or any topology.

Template 6.7 (Ghost discharge). *Assume*

- (H1) *every orbit that never descends from a point outside A eventually enters each set U_h and stays there, and*
- (H2) *every $x \notin A$ has an $h(x)$ and a $K(x)$ with $T^t(x) \notin U_{h(x)}$ for all $t \geq K(x)$.*

Then $\bigcap_k N_k = \emptyset$, so the system is live.

Proof. If x lay in every N_k , its orbit would eventually be inside $U_{h(x)}$ by the first hypothesis and eventually outside it by the second. $\square$

Open obligation. Both hypotheses, for a system of interest. For the Collatz map Γ contains the 2-adic fixed point -1 (Remark 6.5), and both are unproved; for the template to say something about the integers, separation must be established by arguments about the integers alone. What is established is the concrete structure of Section 6.1: descent at step k is the integer inequality $2^{A_k}n > 3^k n + B_k$, and no fixed number of low-order binary digits predicts the exponents that enter it (Corollary 6.4). Orbits of $2^L - 1$ imitate the ghost for $L - 1$ steps.

6.2 G6: Endogenous Needle Generation

Name. A *needle* is an obstruction point; here the run's own history *generates* them.

Let a run on a countable set X be $x_{n+1} = T(x_n, S_n)$, where $S_n = G(x_0, \dots, x_n)$ is an *obstruction set computed from the run's own history*. A point is a *hole* of a finite window $W \subseteq X$ at time n if it lies in W and has not been visited by time n . The intended law says that if the run is driven back towards the holes of a window faster than its history adds obstructions there, every hole is eventually visited; and if the obstructions around a hole grow faster, the hole is permanent. Only the final step of each direction is proved.

Proposition 6.8 (Covering and hole formation). *1. Let $h(n)$ be the number of holes of a window at time n . If for every n with $h(n) > 0$ there is an $m > n$ with $h(m) < h(n)$, then for every n there is an $N \geq n$ with $h(N) = 0$.*

2. If y is unvisited at time N , and for every $n \geq N$ non-visitation of y at time n implies non-visitation at time $n + 1$, then y is unvisited at every time $n \geq N$.

Proof. Part 1 is strong induction on $h(n)$; part 2 is induction on $n - N$. □

The hypothesis of part 2 is essentially its conclusion.

Open obligation. A quantitative comparison, for a system of interest, between how often the run moves into holes and how fast its history adds obstructions, strong enough to imply the premises of Proposition 6.8. Two designed variants of Recamán's rule satisfy those premises directly: with jumps $2n$ instead of n only even values are visited, so every odd number is a permanent hole; with jumps of constant size 1 the run visits exactly $\{0, \dots, n\}$ by time n . For Recamán's sequence itself — $a_n = a_{n-1} - n$ when that is positive and unused, and $a_n = a_{n-1} + n$ otherwise — whether every positive integer appears is open; 852655 remains the smallest missing value in long computations [35, 36]. It is a candidate, not an instance. Here *endogenous* means that the run itself produces the obstruction.

6.3 G9: Defect-Evacuation-to-Transport

Name. Local defects *evacuate* when they stop accumulating and turn into moving patterns, that is, into *transport*.

Let a cellular automaton or other local update act on configurations over a lattice with translations, and fix a reference background. A *defect* is a finite witness that the configuration differs from the background on a local patch. The *defect census* is a ledger that follows every defect from time n to time $n + 1$. Each defect is continued, annihilated, bound into a finite composite, absorbed into a transporter, or marked unresolved. Every new defect is charged to a creation event. A *transporter* is a finite pattern with support W , period $\tau > 0$ and displacement d . It is certified by the recurrence $c_{t_0+(k+1)\tau}(x+d) = c_{t_0+k\tau}(x)$ for x in $W + kd$; that equality, not visual recognition, certifies motion. The transporter is not declared at the start; it appears during the run.

Theorem 6.9 (Census bound). *Let u_n be the number of unresolved defects at time n . If $u_{n+1} \leq u_n$ for all $n \geq N$, then $u_n \leq u_N$ for all $n \geq N$. In particular (u_n) is bounded.*

Proof. Induction on $n \geq N$. □

Template 6.10 (Evacuation into transport). *Assume that the run carries an exact census whose unresolved count eventually stops growing, and*

- (H1) *every defect that stays unresolved long enough is annihilated, bound into a stationary composite, or assigned to one of finitely many transporter records, each satisfying its recurrence.*

If some persistent defect is assigned to a transporter with $d \neq 0$, then from some time on every persistent defect assigned to a transporter is covered by finitely many certified records.

Proof. After the absorption time, the records to which persistent defects are assigned form the required finite family. □

Open obligation. (H1), which carries all the content; the conclusion is then a restatement. For the traffic automaton rule 184 the obligation is partly met. On a ring of $n \geq 2$ cells a *jam* is an occupied cell whose right neighbour is occupied.

Proposition 6.11 (Rule 184, partial results). *On a ring of $n \geq 2$ cells under rule 184:*

1. *a jam-free configuration stays jam-free, and every car advances one cell per step, so from any jam-free time on the run satisfies the recurrence with period 1 and displacement +1;*
2. *the four-cell configuration 1100 has one jam, becomes 1010 after one step, and then alternates between 1010 and 0101;*
3. *the six-cell configuration 001011 has density $\frac{1}{2}$ and one jam, and after one step it still has one jam;*
4. *the number of cars is conserved;*
5. *for $2 \leq n \leq 8$, every configuration with at most $n/2$ cars is jam-free after n steps, and on four cells after one step.*

Proof. Parts (1) to (4) follow from the update rule, a car moving exactly when the cell ahead is empty; part (5) is a finite check. (1) In a jam-free configuration every car has an empty cell ahead and moves into it, and no two cars move into the same cell. (2) and (3) are direct computations. (4) A car leaves a cell only for an empty right neighbour, and a car enters an empty cell only from its left neighbour, so each move removes one car from one cell and adds it to the next. (5) is a finite check over all configurations of each ring. □

Figure 4 shows the runs of parts 2 and 3.

So low-density rings of up to eight cells evacuate their jams within n steps, while the six-cell example shows that the jam count need not fall at every update; evacuation on every ring remains unproved, and so does the assignment of each persistent defect required by (H1). The classical picture is imported: below density $\frac{1}{2}$ blocked cars evacuate and particles move freely, above it the holes do, and at density $\frac{1}{2}$ the configuration settles into the alternating pattern [21, 33, 10]. Langton's ant, whose highway conjecture is open, is not an instance of any claim here; only its unboundedness is proved [8].

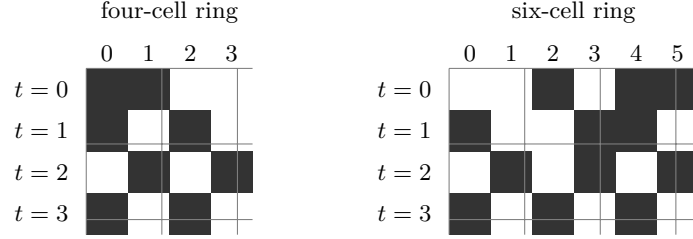


Figure 4. Rule-184 updates on two rings, with time downward, cell indices increasing to the right, and occupied cells filled. The rightmost cell is adjacent to cell 0 in each ring.

6.4 G11: Local-Rule Global-Anti-Symmetry

Name. Local rules force the absence of every translational symmetry, a *global anti-symmetry*.

Let C be a finite set of local rules for configurations $x : \mathbb{Z}^d \rightarrow A$ over a finite alphabet A , for instance a set of Wang tiles. A nonzero $p \in \mathbb{Z}^d$ is a *period* of x when $x(v + p) = x(v)$ for all v , and x is *aperiodic* when it has no nonzero period. A *hierarchy* for C consists of scales $L_k \rightarrow \infty$ and block types at each scale such that every admissible configuration decomposes into blocks at every scale, larger blocks being composed of smaller ones.

Template 6.12 (Hierarchy forces aperiodicity). *Assume*

- (H1) *for every admissible configuration x and every nonzero p , some scale k has block or marker structure in x , forced by the rules of C , that is not invariant under translation by p , and*
- (H2) *whenever forced structure of x at some scale is not invariant under translation by p , the vector p is not a period of x .*

Then every admissible configuration is aperiodic.

Proof. If an admissible x had a nonzero period p , the first hypothesis would give a scale whose forced structure breaks p , and the second would say that p is not a period. $\square$

The mechanism is Robinson’s [41]. Existence of an admissible configuration is a separate input: the template is stated for every admissible configuration and is vacuous if there are none.

Open obligation. A hierarchy for a given tile set, together with a finite region R_p on which the break of each period is visible. Deciding whether a finite Wang tile set admits any tiling of the plane is undecidable [1], so no algorithm sorts an arbitrary tile set into periodic, aperiodic or empty. The classical constructions establish the hierarchy for particular tile sets: Berger’s set, Robinson’s six tiles [41], Penrose’s kites and darts [39], and the hat and spectre monotiles [43, 44].

7 What the laws do not give

Each law is easy to read more strongly than it is stated. Table 4 lists the inferences we expect a reader to be tempted by, and what blocks each one. A complete register of non-implications, with a separating example for each, is in Supplement Section S5.

Table 4. Tempting inferences and what blocks them.

Tempting inference	Why it fails
The ghost-discharge template says something about the Collatz conjecture.	Both of its hypotheses are unproved for the Collatz map (Section 6.1). The split pairs of Theorem 6.3 rule out fixed-depth prediction; they are not evidence about liveness.
A decreasing escrow may decrease before or after a change of presentation.	The comparison must cover the step and the change of presentation together (Section 5.1).
A statistic that tracks cost is a potential.	Only a nonnegative potential entering the telescoping identity bounds cost (Theorem 4.1).
Covering every target so far shows the cover is exhaustive.	Exhaustion is the hypothesis of Theorem 4.3; for Erdős–Straus it is the open problem.
A closed target-free family for base 2 suggests one for 196 in base 10.	None is known, and the base-2 family was found by hand for one orbit (Section 4.3).
A long run without reaching 852655 shows it is a permanent hole of Recamán’s sequence.	A finite computation says nothing about later steps; Recamán’s sequence is a candidate, not an instance (Section 6.2).
Return pressure exceeding obstruction growth forces coverage.	The implication from the rate conditions to the premises of Proposition 6.8 is open (Section 6.2).
The certificate theorem for the angel game finds winning strategies.	It checks given strategies (Theorem 5.2); the thresholds are imported.
Confluent move systems have route-independent counters.	Example 4.13 is confluent and its counter vectors differ.
Any terminating move system has a unique final configuration.	Example 4.14 terminates without one.
Visible motion of a pattern is transport.	Only a checked recurrence with displacement certifies it (Section 6.3); Langton’s ant is not an instance.
Persistence for all iterates can be observed.	It must be proved from protection and regeneration (Theorem 4.16); a finite run gives bounded evidence only.
Local rules whose finite patches extend always force aperiodicity.	A hierarchy must be supplied (Template 6.12), and whether a tile set tiles at all is undecidable.
Every impossible global colouring has a finite witness.	Only under the compactness hypothesis (Proposition 4.22), which uses the axiom of choice.
Passing every congruence condition means an integer occurs in the orbit.	Reciprocity obstructions are missed despite passing (Section 4.7).

8 Formalization

What is checked. The general statements of all thirteen entries are checked in Lean 4 [15], in a library with no external dependencies, together with the Collatz ledger and split pairs (Theorems 6.2 and 6.3), the four-phase family and the orbit of 22 (Theorem 4.6 and Corollary 4.7), Ducci nilpotency (Theorem 4.18), the two separating examples for G8, and the instances in Examples 4.4, 4.17 and 4.24 and Proposition 6.11, the binary counter of Example 4.2, a three-site sandpile, the colouring obstruction of the Moser spindle and the integer Ducci bound for $n = 4$. The last column of Table 1 says, entry by entry, what is covered; sandpiles on arbitrary graphs, the plane embedding of the Moser spindle, the other remarks and the imported results are proved on paper only. Some Lean statements are formally weaker than the printed ones, for example Theorem 5.1 without the extraction of a terminal stage, and some carry a hypothesis they do not use; both are listed in Supplement Section S7.

Axioms. The library builds without `sorry` or added axioms and uses at most the standard axioms `propext`, `Quot.sound` and `Classical.choice`.

Where to look. Following Pollack [40], what a reader must trust is the kernel and the statements. Supplement Section S2 maps every numbered result of this paper to its Lean declarations, file and line; Supplement Section S7 gives the axioms of each declaration and a clause-by-clause comparison with the printed statements; Supplement Section S9 gives the commands that rebuild the library.

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